

Subject: Re: CATCH manuscript

From: Ang Li <ang.li@psych.ox.ac.uk>

Date: 9/8/26, 07:28

To: Jian Zeng <j.zeng@imb.uq.edu.au>, Peter Allen <p.allen@garvan.org.au>

Sorry forgot mention that

- 157 cell types with valid conLDSC, scDRS, CATCH, and BH-FDR results.
- Six ambiguous CD4/CD8 subtypes retained with scDRS P values, but NA for conLDSC, CATCH, and FDR because CELLECT collapsed their labels.

CELLECT has bugs in identifying cell type names.

Thank you.

Best,

Ang

From: Ang Li <ang.li@psych.ox.ac.uk>

Date: Monday, 7 September 2026 at 22:09

To: Jian Zeng <j.zeng@imb.uq.edu.au>, Peter Allen <p.allen@garvan.org.au>

Subject: Re: CATCH manuscript

[Test_CATCH](#)

TS_pc h5ad with 163 cell types is big , link in above one drive. Please let me know if you cannot access . I also put Cepo, cov for scDRS and CATCH result in attached as well.

FYI: CATCH results based on atlas data TS-pc and T2D_06 GWAS is
T2D_06_TabulaSapiens_pc_minCell20_catch_combined_Ang_163_cell_types.tsv .

Thank you.

Best,

Ang

From: Jian Zeng <j.zeng@imb.uq.edu.au>

Date: Monday, 7 September 2026 at 08:12

To: Peter Allen <p.allen@garvan.org.au>

Cc: Ang Li <ang.li@psych.ox.ac.uk>

Subject: Re: CATCH manuscript

Was it using a single CPU? Can the process be paralleled?

I'm surprised by the need of 39 hours which is quite long. What are the most time consuming parts?

Ang, can you share the human atlas data? We may not need to provide the raw data if they are directly used by scDRS but should we provide LDSC annotation on our website?

Best regards,
JZ

--

Dr Jian Zeng

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Personal Zoom: <https://uqz.zoom.us/j/6544643277>

From: Peter Allen <p.allen@garvan.org.au>
Date: Monday, 7 September 2026 at 4:57 PM
To: Jian Zeng <j.zeng@imb.uq.edu.au>
Cc: Ang Li <ang.li@psych.ox.ac.uk>
Subject: Re: CATCH manuscript

Hi Jian,

This is for one trait (T2D) and the Fasolino, 2022 dataset (47k cells) was what I used. Computing hours is 39 for this trait (not sequential as many of the jobs run concurrently). I'm happy to run another dataset through if you wouldn't mind sending me the link to it and what trait you would like to test -- another example to ensure reproducibility between Ang's and my pipeline would be great.
Peter

On 9/7/26 16:48, Jian Zeng wrote:

Is 39 hours for all traits or for one trait? We should report this for one trait.

I would prefer to use human atlas dataset analysis as the example because this is what we recommend.

Best regards,
JZ

--

Dr Jian Zeng

Group Leader
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Re: CATCH manuscript

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From: Peter Allen <p.allen@garvan.org.au>
Date: Monday, 7 September 2026 at 4:42 PM
To: Jian Zeng <j.zeng@imb.uq.edu.au>
Cc: Ang Li <ang.li@psych.ox.ac.uk>
Subject: Re: CATCH manuscript

Hi Jian,

Computing hours (from the updated run): 39 hours

Memory: 25GB

I think for transparency, mentioning the dataset size (~47k cells) helps the reader understand relative times for Cepo (arguably the longest process).

Best regards,

Peter

Peter C. Allen, PhD



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On 9/7/26 16:25, Jian Zeng wrote:

Thanks, Peter.

For this sentence in the main text, what would be your suggestion:

For a standard CATCH analysis with ~1M SNPs in the GWAS summary statistics and 20,000 protein-coding genes, it took approximately ## computing hours and ## GB memory on a standard high-performance computing cluster using our Nextflow pipeline (<https://github.com/PeterCAllen/CATCH-pipeline>).

Should we also mention the sc dataset size?

Best regards,
JZ

--

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From: Peter Allen <p.allen@garvan.org.au>

Date: Monday, 7 September 2026 at 3:20 PM

To: Ang Li <ang.li@psych.ox.ac.uk>; Jian Zeng <j.zeng@imb.uq.edu.au>

Subject: Re: CATCH manuscript

Hi Ang,

Thanks for sending this. Largely our parameters are similar: We use GCTA's defaults (`--diff-freq 0.2 and --mBAT-svd-gamma 0.9`) and --ld-window 1 & --mBAT-window 10. We do use baselineLD v1.2.

The big difference and probably the source of variation is the fact that we subset mBAT genes to just the protein-coding (19k genes vs your 58k). For the mBAT LD ref panel, we used the same 1000G EUR panel as conLDSC (1000G.EUR.QC, 489 samples, 779k snps), which I'll switch it to match your MAGMA ref.

Once I've got the results from the recent run, I'll attach and send.

All the best,
Peter

Peter C. Allen, PhD



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From: Ang Li <ang.li@psych.ox.ac.uk>

Sent: Monday, September 7, 2026 1:13 PM

To: Peter Allen <p.allen@garvan.org.au>; Jian Zeng <j.zeng@imb.uq.edu.au>

Subject: Re: CATCH manuscript

Hi Peter,

Thanks for your rapid response! !

I attached my results in the same format as yours and some figures.

For the biological conclusion: reasonably acceptable. Both analyses identify exactly the same three cell types at BH FDR <0.05: pancreatic D, pancreatic beta, and unknown. CATCH rank correlation is 0.976; conLDSC is 0.964 and scDRS is 0.891.

For exact computational replication: not that the same. So I think their maybe some upstream differences I listed that could be checked:

- **mBAT window:** --mBAT-wind 10, meaning approximately ± 10 kb around each gene.
- **mBAT gene coordinates:** geneMatrix.hg19.loc.mBATversion.txt(attached)
- **mBAT LD reference:** 1000 Genomes European: g1000_eur_bfile/g1000_eur. (Download from <https://cncr.nl/research/magma/>)
- **Other mBAT settings:** --diff-freq 0.2 and --mBAT-svd-gamma 0.9.
- **GCTA version:** GCTA 1.94.1.
- **conLDSC baseline:** baseline-LD **v1.1**, if you use v1.2 it is ok and

good.

- **conLDSC gene-to-SNP window: ± 100 kb.**
- **LDSC LD-score window: `--ld-wind-cm 1`, meaning 1 centimorgan, not 1 kb.**

My conLDSC all from CELLECT set, so I think LD ref should be the same as yours.

Thank you!
Best,
Ang

From: Peter Allen <p.allen@garvan.org.au>

Date: Monday, 7 September 2026 at 03:04

To: Jian Zeng <j.zeng@imb.uq.edu.au>, Ang Li <ang.li@psych.ox.ac.uk>

Subject: Re: CATCH manuscript

Happy Monday Ang and Jian,

I've completed the update for the CATCH pipeline with the two methods:

- ConLDSC-Cepo: cell-type-specificity (Cepo) + stratified LD score regression-
cts (updated python3)
- scDRS (mBAT-combo based)

Claude gave a summary of the pipeline:

- Takes a single-cell reference (h5ad) and GWAS summary statistics as input; outputs one p-value per cell type (CATCH_P) indicating trait relevance, plus the two component p-values for transparency.
- Accepts GWAS in whatever format you already have it — raw sumstats, GCTA-COJO (.ma), or pre-munged LDSC (.sumstats.gz) — no manual reformatting needed.
- Fully containerized (Singularity) and implemented as a Nextflow pipeline, so it's portable across HPC clusters and reproducible end-to-end from raw inputs to final result.
- Recently migrated off CELLECT's Python 2 LDSC dependency to the actively maintained Python 3 fork, and simplified from 4 to the 2 methods actually used in the current manuscript draft (the other two, conLDSC-GES and seismic-mBAT-combo, remain available as optional validation branches).
- Validated end-to-end on real data (T2D GWAS + pancreatic islet scRNA-seq): recovered biologically expected top hits (beta and delta cells for a T2D trait) and Cepo output correlates $r \approx 0.999$ with independently computed reference scores.

Re: CATCH manuscript

- Typical resource footprint on this dataset: ~31 CPU-hours, ~25GB peak memory (dominated by the Cepo permutation step).

Quick results summary from the Fasolino, 2022 T2D run (results attached):

- CATCH Cauchy combination of conLDSC-Cepo + scDRS across all 10 pancreatic cell types. Top hits are:
 - o pancreatic_D_cell (CATCH_P=0.0015)
 - o type_B_pancreatic_cell (CATCH_P=0.0032) followed by unknown (0.0094).
- Comparing Cepo scores, our pipeline's Cepo output vs your precomputed scores had a pearson $r=0.999$ with per cell-type ranging from $r=0.997$ - 0.999 .

Re: running locally: mostly yes (NORMALIZE_H5AD ~4.5GB, scDRS steps ~3–3.5GB, per-chromosome mBAT/conLDSC steps ~1–2GB each), but the Cepo run topped out ~25gb (100 permutations across 47k cells).

Fingers crossed these results match what you've found.

All the best,
Peter

Peter C. Allen, PhD



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From: Peter Allen p.allen@garvan.org.au

Sent: Sunday, September 6, 2026 9:58 AM

To: Jian Zeng j.zeng@imb.uq.edu.au; Ang Li ang.li@psych.ox.ac.uk

Subject: Re: CATCH manuscript

Hi Jian,

As it currently stands, HPC is the optimal env with concurrent processes sent as individual jobs on the cluster. I may posit to Claude to determine if we can optimize this for local machines with 16gb of memory.

Peter

From: Jian Zeng <j.zeng@imb.uq.edu.au>
Sent: Sunday, 06 September 2026 09:46:10
To: Peter Allen <p.allen@garvan.org.au>; Ang Li <ang.li@psych.ox.ac.uk>
Subject: Re: CATCH manuscript

Thank you very much, Peter.

Just curious, is it possible to run on a laptop or we require a HPC environment?

JZ

--

Dr Jian Zeng

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From: Peter Allen <p.allen@garvan.org.au>
Date: Sunday, 6 September 2026 at 9:16 AM
To: Ang Li <ang.li@psych.ox.ac.uk>
Cc: Jian Zeng <j.zeng@imb.uq.edu.au>
Subject: Re: CATCH manuscript

Hi Ang,

Thank you for sending this. I started updating the pipeline Friday and am working through testing with your data. I'll be sure to send you the results and update the compute hours and memory as soon as it's finished.

All the best,
Peter

From: Ang Li <ang.li@psych.ox.ac.uk>
Sent: Sunday, 06 September 2026 08:29:37
To: Peter Allen <p.allen@garvan.org.au>
Cc: Jian Zeng <j.zeng@imb.uq.edu.au>
Subject: Re: CATCH manuscript

Hi Peter,
We settled the final manuscript as attached for your information and leave the blank in line 394 for your comments.
If you need more time to check please let me know. Thank you! 😊
Best,
Ang

From: Ang Li <ang.li@psych.ox.ac.uk>
Date: Thursday, 3 September 2026 at 11:57
To: Peter Allen <p.allen@garvan.org.au>
Cc: Jian Zeng <j.zeng@imb.uq.edu.au>
Subject: Re: CATCH manuscript

Hi Peter,
Thank you for your effective work! !
Sorry for my delay since we added some more simulations afterwards.
We have updated the ms after receiving comments from collaborators. And ended in CATCH using conLDSC Cepo and mBAT-combo based scDRS. Could you please check in the Fig6 attached for pipeline that yours would be like. And the test input with real GWAS of T2D(cojo format and ldsc format) and pancreatic scRNAseq is in the link for you.
<https://drive.google.com/drive/folders/1rNr9gWp-OYmAfR5XTXOwOF0jyQHvYbii?usp=sharing> Also I put Cepo score from the scdata as well for you to check the consistency.

And we would like to highlight in the main: " it took approximately ## computing hours and ## GB memory on a standard high-performance computing cluster using our Nextflow pipeline (<https://github.com/PeterCAllen/CATCH-pipeline>)."

Also now <https://github.com/bulik/LDSC> LDSC recently has updated the python3 version, do you think it would be possible for us to update with this new LDSC? Now I think in your nextflow is still based CELLECT version, which I think CELLECT is not that user friendly. We debuted the source code of CELLECT in our real analysis several times.

Please let me know if I missed any files for you.
Thank you!!

Best,
Ang

From: Peter Allen <p.allen@garvan.org.au>

Date: Friday, 7 August 2026 at 00:13

To: Ang Li <ang.li@psych.ox.ac.uk>

Subject: Re: CATCH manuscript

Ah, too easy then! No worries at all :-)

Peter

On Aug 7 2026, at 9:06 am, Ang Li <ang.li@psych.ox.ac.uk> wrote:

Hi Peter,

Sorry for the confusion.

Only the authors of the 1st version of the ms need to sign I think.

I am preparing to send you the data test tomorrow. I just come back from holiday 😊. Many thanks for your so rapid response.

Best,

Ang

On 6 Aug 2026, at 23:28, Peter Allen

<p.allen@garvan.org.au> wrote:

Hi Ang,

I'm not on this form, but happy to sign.

Have you been able to take a look at the next flow pipeline?

// Peter

From: Ang Li <ang.li@psych.ox.ac.uk>

Sent: Friday, 07 August 2026 07:26:37

To: Jian Zeng <j.zeng@imb.uq.edu.au>; Tian Lin

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Re: CATCH manuscript

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Subject: Re: CATCH manuscript

Dear All,

Apologies for the typo in the document I sent in my previous email. **Please use the document in this email.**

We have added eight additional authors to the revised manuscript.

For the authors on the first submission to approve the addition of the new authors:

Per Nature requirements, could you please approve the addition of these authors with your Signature in **Section 6**. And could you please sign in the form attached with your **Signature + Date** and send back to all? Thank you!
(Attached:PRE_Authorship_form_springer_nature_pre-acc_AL_06082026.pdf)

And for **Section 5**, could you please check the New Acknowledgements and New Disclosure if you would like to add any additional information. Please just add and send to us all.

Thank you very much again for your contributions and support! 😊

Best,
Ang

From: Jian Zeng <j.zeng@imb.uq.edu.au>
Date: Monday, 3 August 2026 at 08:01
To: Peter Allen <p.allen@garvan.org.au>, Yong Wang <yong.wang@psych.ox.ac.uk>, Haochen Cui <haochen.cui@kellogg.ox.ac.uk>, Tian Lin <tian.lin@imb.uq.edu.au>, Yuliangzi Sun <yuliangzi.sun@uq.edu.au>, Xuemin Wang <x.wang20@uq.edu.au>, Xiao Tan <xiao.tan@uq.edu.au>, Alicia Walker <alicia.walker@psych.ox.ac.uk>, Siqi Wang <siqi.wang@ndph.ox.ac.uk>, Zheng Yao <zheng.yao1@student.uq.edu.au>, Ruolan Zhao <r.zhao1@uq.edu.au>, jian.yang <jian.yang@westlake.edu.cn>, shuyang.yao@ki.se <shuyang.yao@ki.se>, patrick_sullivan@med.unc.edu <patrick_sullivan@med.unc.edu>, jens.hjerling-leffler@ki.se <jens.hjerling-leffler@ki.se>
Cc: Ang Li <ang.li@psych.ox.ac.uk>, Naomi Wray <naomi.wray@psych.ox.ac.uk>
Subject: CATCH manuscript

Dear All,

We are pleased to share the revised version of our manuscript on benchmarking GWAS and scRNA-seq integrative methods.

Nature Methods initially rejected the manuscript. We appealed the decision, and fortunately the editors agreed to consider a substantially revised manuscript. This has therefore been a major revision, involving substantial restructuring and rewriting of the manuscript, many additional analyses, and new figures and tables.

Because the revision is so extensive, we have not marked the edits in the manuscript, as nearly every section has been substantially revised. We hope this makes the manuscript easier to read.

Ang has devoted a tremendous amount of time and

Re: CATCH manuscript

effort to leading the revision and responding to the reviewers' comments. At this stage, we are not looking for suggestions for additional analyses, as we believe the manuscript is already comprehensive and would like to avoid delaying the resubmission. Instead, we would greatly appreciate any comments on the interpretation, presentation, clarity, and wording of both the manuscript and the rebuttal letter.

We are aiming to resubmit the revision to Nature Methods within the next two weeks. We understand that some of you may be on holiday at the moment, and we apologize for the inconvenience. However, we would greatly appreciate if you could send us your comments by **17 August**.

We have added several co-authors in the revised version because their significant contributions to the new analyses, software development, and manuscript revisions.

Finally, please check your affiliations and let us know if you would like us to include any funding acknowledgments.

Thank you very much again for your support throughout this project. We really appreciate all your contributions.

Best regards,
Jian, Ang, and Naomi

--

Dr Jian Zeng

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Personal Zoom: <https://uqz.zoom.us/j/6544643277>